

AI BIAS AUDITOR v3.0

USER GUIDE - RINGKAS

Panduan Penggunaan dan Interpretasi Metrics

Audit Teknologi dan Sistem Informasi

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1. GLOSSARY - DEFINISI ISTILAH

Approval Rate

Persentase applicants yang disetujui (loan approved) dari total applicants dalam suatu kelompok. Formula: $(\text{Jumlah Approved} / \text{Total Applicants}) \times 100\%$. Contoh: Jika 72 dari 100 applicants disetujui, approval rate = 72%.

Credit Score

Nilai numerik (skala 300-850) yang merepresentasikan kelayakan kredit seseorang. Semakin tinggi skor, semakin rendah risiko gagal bayar. Kategori: 300-500 (High Risk), 500-600 (Medium Risk), 600-700 (Low Risk), 700-850 (Very Low Risk).

Age Group

Klasifikasi applicants berdasarkan umur: Young (≤ 45 tahun) dan Old (> 45 tahun). Threshold 45 tahun digunakan untuk mendeteksi age discrimination (ageism).

Bias (Bias Sistemik)

Perlakuan tidak adil secara sistematis terhadap kelompok tertentu dalam proses pengambilan keputusan AI. Dalam konteks ini: diskriminasi berdasarkan umur (age discrimination).

Disparate Impact (DI) Ratio

Rasio approval rate kelompok minoritas dibanding kelompok mayoritas. Formula: $(\text{Approval Rate Old} / \text{Approval Rate Young})$. Standard legal: DI Ratio ≥ 0.80 (80% Rule dari EEOC). Jika < 0.80 , indikasi diskriminasi yang dapat digugat secara hukum.

p-value

Probabilitas bahwa perbedaan yang diamati terjadi karena kebetulan (random chance). Interpretasi: $p < 0.05$ = perbedaan signifikan secara statistik (bukan kebetulan). $p < 0.01$ = sangat signifikan. $p > 0.05$ = tidak ada perbedaan yang signifikan.

2. CARA BACA METRICS

2.1. Score Gap (Selisih Skor)

Perbedaan rata-rata credit score antara kelompok Young dan Old. Formula: Avg Score Young - Avg Score Old.

Interpretasi:

- Gap < 50 points: Perbedaan kecil, mungkin wajar
- Gap 50-100 points: Perbedaan moderat, perlu investigasi
- Gap > 100 points: Perbedaan besar, indikasi BIAS KUAT

Contoh: Jika Young avg = 632 dan Old avg = 439, maka Gap = 193 points (BIAS!).

2.2. Approval Gap (Selisih Approval Rate)

Perbedaan approval rate antara kelompok Young dan Old. Formula: Approval Rate Young - Approval Rate Old.

Interpretasi:

- Gap < 10%: Perbedaan kecil, relatif fair
- Gap 10-20%: Perbedaan moderat, perlu monitoring
- Gap > 20%: Perbedaan besar, indikasi BIAS

Contoh: Jika Young = 72.4% dan Old = 28.6%, maka Gap = 43.8% (BIAS SIGNIFIKAN!).

2.3. T-Test dan Mann-Whitney U Test

Statistical tests untuk menguji apakah perbedaan antara 2 kelompok signifikan.

T-Test: Parametric test (asumsi distribusi normal)

Mann-Whitney U: Non-parametric test (tidak asumsi distribusi)

Cara Baca p-value:

- $p < 0.001$: Extremely significant (99.9% confidence)
- $p < 0.01$: Highly significant (99% confidence)
- $p < 0.05$: Significant (95% confidence) - THRESHOLD STANDARD
- $p \geq 0.05$: Not significant (no strong evidence of difference)

Contoh: p-value = 0.00000001 artinya perbedaan SANGAT SIGNIFIKAN, bukan kebetulan.

2.4. Cohen's d (Effect Size)

Ukuran seberapa BESAR perbedaan antara 2 kelompok, independent dari sample size. Formula: $(\text{Mean1} - \text{Mean2}) / \text{Pooled Standard Deviation}$.

Interpretasi (Cohen, 1988):

- $d = 0.2$: Small effect (perbedaan kecil)
- $d = 0.5$: Medium effect (perbedaan sedang)
- $d = 0.8$: Large effect (perbedaan besar)
- $d > 1.0$: Very large effect (perbedaan sangat besar)

Contoh: Cohen's $d = 3.142$ artinya LARGE EFFECT, bias sangat besar dan praktis signifikan.

2.5. Disparate Impact Ratio (80% Rule)

Legal standard dari EEOC (Equal Employment Opportunity Commission) untuk mendeteksi diskriminasi.

Formula: $(\text{Selection Rate Minoritas} / \text{Selection Rate Mayoritas})$

Standard Legal:

- $DI \geq 0.80$: PASS (Compliant, fair)
- $DI < 0.80$: FAIL (Non-compliant, potential legal violation)
- $DI < 0.50$: Severe violation (highly discriminatory)

Contoh: $DI = 0.394$ artinya Old group hanya 39.4% kemungkinan approved dibanding Young group. Ini FAIL karena < 0.80 , indikasi DISKRIMINASI yang dapat digugat.

3. INTERPRETASI HASIL AUDIT

3.1. Kapan Bias Terdeteksi?

Bias dinyatakan TERDETEKSI jika memenuhi SALAH SATU kriteria berikut:

- Score Gap > 100 points
- Approval Gap > 20%
- Disparate Impact Ratio < 0.80
- T-Test p-value < 0.05 (signifikan)
- Cohen's d > 0.8 (large effect)

Jika SEMUA kriteria terpenuhi: BIAS KUAT dan JELAS (immediate action required).

3.2. Level Severity (Tingkat Keparahan)

LOW SEVERITY:

- Score Gap: 50-100 points
- Approval Gap: 10-20%
- DI Ratio: 0.70-0.79
- Rekomendasi: Monitoring dan minor adjustment

MEDIUM SEVERITY:

- Score Gap: 100-150 points
- Approval Gap: 20-30%
- DI Ratio: 0.50-0.69
- Rekomendasi: Model review dan fairness intervention

HIGH SEVERITY:

- Score Gap: > 150 points
- Approval Gap: > 30%
- DI Ratio: < 0.50
- Rekomendasi: Immediate model redesign, legal review

3.3. Compliance Status

COMPLIANT: DI Ratio $\geq$ 0.80, no significant bias detected, meets fairness criteria.

NON-COMPLIANT: DI Ratio < 0.80, significant bias detected, violates 80% Rule.

- Risk: Legal liability, regulatory fines, reputational damage
- Action: Immediate intervention required

4. REFERENSI & TINJAUAN PUSTAKA

4.1. Legal & Regulatory Standards

- EEOC (1978). Uniform Guidelines on Employee Selection Procedures. 80% Rule for Disparate Impact.
- UU No. 8/1999 tentang Perlindungan Konsumen (Indonesia).
- OJK Regulation on Financial Services Consumer Protection.
- ISO/IEC 24027:2021 - Bias in AI systems and AI aided decisions.

4.2. Academic References

- Barocas, S., & Selbst, A. D. (2016). Big Data's Disparate Impact. California Law Review, 104, 671.
- Feldman, M., et al. (2015). Certifying and Removing Disparate Impact. ACM SIGKDD.
- Mehrabi, N., et al. (2021). A Survey on Bias and Fairness in Machine Learning. ACM Computing Surveys.
- Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.).

4.3. Tools & Libraries

- Streamlit 1.32+ - Interactive web dashboard framework
- Pandas 2.2+ - Data manipulation and analysis
- SciPy 1.11+ - Statistical computing and hypothesis testing
- Plotly 5.19+ - Interactive data visualization
- Scikit-learn 1.7+ - Machine learning metrics and evaluation